FUTURE PROJECT BRIEF • 2026-05

Astronomy's Computation Crisis

Top 10 problems. The AI / kernel solutions. Where to build first.

"The instruments are done. The substrate isn't."

Companion document to the **Argus** project — a differentiable atmospheric retrieval kernel

`~/Desktop/Future/Argus`

The setup — May 2026

Vera C. Rubin Observatory — live since Feb 2026.
10 TB / night. Up to **10 million alerts / night**. 60-second latency to public stream.

Square Kilometre Array — early science 2026, full ops 2028.
~**1 EB / year** archived. 100–1000× that before correlation.

JWST + Ariel (2029) — native-resolution exoplanet spectra. Retrievals can't keep up; data is binned away.

LIGO O5 + LISA (2035) + ET — ~100k GW detections / year by 2030s. Global fit on overlapping signals: open problem.

The instruments are done. The substrate isn't.

The 10 problems at a glance

#	Problem	Pain (one phrase)
1	LSST alert firehose	10M alerts/night, 60s, 7 brokers
2	SKA visibility stream	1 EB/yr archived, EB/day uncorrelated
3	Radio interferometric imaging	inversion dominates at SKA fields
4	Cosmological N-body	millions of core-hours per run
5	GRMHD black-hole simulations	80M CPU-hr for the EHT library
6	GW parameter estimation	exponential cost; LISA global fit unsolved
7	**Exoplanet atmospheric retrieval**	JWST data binned away to make MCMC tractable
8	Petabyte catalog cross-match	Gaia + LSST + Euclid + Roman, no shared join
9	Strong gravitational lens modeling	MCMC hours/system, 100k Euclid lenses incoming
10	Multi-messenger coordination	seconds-to-minutes; today: GCN circulars + Slack

#7 highlighted as the chosen Argus wedge — explained at the end.

#1 — LSST Alert Firehose

PROBLEM #1 • Real-time transient classification at Rubin scale **TIME-PERISHABLE**

10 TB per night, raw images	**10M** alerts / night peak	**60 s** latency to public stream

Why it hurts. Seven community brokers (Fink, ALERCE, ANTARES, AMPEL, Lasair, Pitt-Google, Babamul) each ingest the full stream and re-implement feature extraction, light-curve embedding, contextual cross-match, classifier inference. No shared feature store. Latency budgets blown by Python glue. Missed kilonovae are missed forever.

AI / algorithm. Transformer encoders over irregular multi-band light curves, self-supervised contrastive embeddings (AstroCLIP / AstroPT), early-classification heads with abstention, active-learning loops driving spectroscopic follow-up.

Kernel opportunity. A C++/CUDA streaming-inference runtime with a stable photometric feature ABI — ONNX Runtime for time-series astronomy. Brokers become *configurations*, not bespoke services.

#2 — SKA Visibility Stream

PROBLEM #2 • Real-time radio data deluge at exascale **HARDWARE-BOUND**

~1 EB/yr archived at Regional Centres	**~1 EB/day** at antennas, pre-correlation	**millions** servers if built today

Why it hurts. Real-time RFI excision, channelization, calibration, and visibility compression must run at the central signal processor. ASKAPsoft, MeerKAT pipeline, LOFAR Cobalt, SDP — each rolled their own. No `libvis` for the world.

AI / algorithm. Learned RFI classifiers replacing hand-tuned masks; neural calibration of antenna gains; learned lossy visibility compression; deep image priors for self-cal.

Kernel opportunity. Streaming SIMD/CUDA visibility kernel, GPU-resident, with a typed visibility IR. Hardest of the ten to validate without a partner facility.

#3 — Radio Interferometric Imaging

PROBLEM #3 • CLEAN, gridding, and inversion at SKA fields

KERNEL-READY

TB-scale per pointing, post-correlation	**Inversion** dominates, not CLEAN	**4+ codes** CASA, WSClean, DDFacet, Cotton-Schwab

Why it hurts. Once visibilities are calibrated, gridding + inverse FFT + deconvolution produces images. At SKA-sized fields, the inversion/prediction step dominates — not CLEAN. Existing pipelines are CPU-first and fragmented across packages.

AI / algorithm. Differentiable interferometric forward models (gridder + FFT + PSF as autograd ops); plug-and-play priors; learned regularizers replacing handcrafted scale choices; neural deconvolution with calibrated uncertainty.

Kernel opportunity. A GPU-resident, differentiable interferometry library — `torch.fft` for radio astronomy. Whoever ships this owns the SKA-era imaging substrate.

#4 — Cosmological N-body Simulations

PROBLEM #4 • Trillion-particle dark-matter precision cosmology

COMPUTE-BOUND

10^{12} particles state-of-the-art runs	**millions** core-hours per run	**TB / PB** memory / snapshots

Why it hurts. Euclid, Rubin, DESI, Roman (~$10B combined) need *thousands* of runs to marginalize over cosmological parameters. Each run takes weeks. FFT and tree-walks dominate. Codes (Gadget-4, ChaNGa, HACC, GIZMO, Arepo) each reimplement gravity solvers; none are differentiable end-to-end.

AI / algorithm. Neural N-body emulators (CAMELS, Lagrangian Deep Learning, Mass-Conditioned Halo Finders) replace runs at <1% cost; differentiable particle-mesh codes (PMWD, JaxPM) enable gradient-based parameter inference; learned super-resolution upgrades coarse runs.

Kernel opportunity. A differentiable, GPU-native N-body kernel with pluggable physics passes. Training emulators needs cluster hours — compute-bound, not idea-bound.

#5 — GRMHD Black Hole Simulations

PROBLEM #5 • Accretion-flow simulation libraries for the EHT

ACCESS-BOUND

80M CPU-hr EHT library on Frontera	**23 TB** the library itself	**8+ codes** BHAC, KORAL, IllinoisGRMHD, HARM…

Why it hurts. Each new EHT target (Sgr A*, M87*, future ngEHT sources) demands a fresh library because magnetic-field topology and spin priors change. Each library run is multi-week, single-physics, and lives in a code that doesn't talk to the next code.

AI / algorithm. Physics-informed neural surrogates trained on the existing library to interpolate accretion-flow parameters in milliseconds; diffusion models conditioned on (spin, accretion mode) producing synthetic image cubes; differentiable GRMHD for accretion-disk parameter inference.

Kernel opportunity. GPU-native relativistic-fluid kernel wrapped in autograd. Library-access negotiation with EHT collab is the gating risk, not the technology.

#6 — Gravitational Wave Parameter Estimation

PROBLEM #6 • Posterior inference for compact-binary coalescences

POLITICAL-BOUND

exp(N) matched-filter cost in N parameters	**hours–days** Bayesian PE per event	**~100k/yr** events expected from CE/ET

Why it hurts. Matched filtering scales exponentially with parameters. **LISA's "global fit"** — jointly inferring parameters of *thousands* of overlapping signals — is the open hard problem of GW astronomy. Production pipeline (LALSuite, IMRPhenom, EOB, NRSur) is C/Fortran with no autograd path.

AI / algorithm. Simulation-based inference via normalizing flows: DINGO has demonstrated <1 second amortized inference vs. days for classical MCMC, with calibrated posteriors. Deep filtering replacing matched filtering. Differentiable waveform models (jaxGW).

Kernel opportunity. A differentiable waveform library + amortized SBI engine. Technically buildable; LSC controls the production pipeline, so adoption is a political wedge.

#7 — Exoplanet Atmospheric Retrieval ★

PROBLEM #7 • Inferring atmospheres from JWST/Ariel spectra

ARGUS WEDGE

seconds–min per RT forward call	10^6–10^7 calls per MCMC retrieval	**native resolution** currently binned away

Why it hurts. JWST and Ariel produce native-resolution spectra. Current retrieval codes (TauREx, POSEIDON, CHIMERA, petitRADTRANS) **bin spectra down** to make MCMC tractable — losing the very signal that argued for big telescopes. Every code is a different scientist's PhD; there is no `libRT`.

AI / algorithm. Differentiable radiative transfer (ExoJAX2 ships this in JAX); neural emulators of opacity tables and chemistry; SBI replacing MCMC; transformer-based posterior estimators.

Kernel opportunity. Differentiable, GPU-native RT kernel with line-by-line opacity, scattering, and chemistry as composable autograd passes. Per-planet datasets are MB — single-GPU MVP. **This is what Argus ships first.**

#8 — Petabyte Catalog Cross-Match

PROBLEM #8 • Spatial joins across the world's surveys

TABLESTAKES

20B LSST sources	**10B** Euclid sources	**6+ surveys** Gaia, WISE, ZTF, PS1, LSST, Euclid, Roman

Why it hurts. Probabilistic positional + photometric + astrometric matching is now a serious distributed-systems problem. Every survey rolls its own join (TOPCAT, X-Match, ARI-Gaia, lsdb). HATS/hipscat is converging; no GPU-native engine exists.

AI / algorithm. Learned probabilistic matchers (positional + colour priors); GNN entity resolution; learned hash families for HEALPix-aware spatial joins.

Kernel opportunity. Columnar HEALPix-indexed catalog format + GPU spatial-join kernel. "DuckDB for the sky." Quietly the highest-leverage primitive — every other domain depends on it.

#9 — Strong Gravitational Lens Modeling

PROBLEM #9 • Reconstructing source + mass from arcs

KERNEL-READY

~100k lenses Euclid will discover	**hours–days** MCMC per system	$10^7\times$ neural-net speedup demonstrated

Why it hurts. Classical MCMC (Lenstronomy, GLEE, GLAFIC) takes hours-to-days per system. JWST adds resolved kinematics, doubling parameter space. The science can't keep up with discovery.

AI / algorithm. CNN/transformer parameter regressors (LEMON, SKiNN); conditional normalizing flows for full posteriors; differentiable ray-tracers + SBI; deep ensembles for outlier triage.

Kernel opportunity. A differentiable lensing kernel callable from any inference engine. Same primitives pattern as exoplanet RT — proves Argus IR generalizes.

#10 — Multi-Messenger Coordination

PROBLEM #10 • Cross-facility alerts in seconds

POLITICAL-BOUND

seconds–min required latency	**GCN + Slack** current state of the art	**RADAR** first AI cross-facility framework, 2026

Why it hurts. GW170817 was the proof of concept. Rubin + LIGO/Virgo/KAGRA + IceCube + Fermi/Swift + ngVLA need to coordinate within minutes when a kilonova fires. Today: GCN circulars, bespoke Slack channels, manual telescope retargeting.

AI / algorithm. Federated streaming inference (compute moves to data); shared event-schema embeddings; ML triggers for telescope retargeting; joint posterior fusion across messengers.

Kernel opportunity. Shared event bus + on-site inference runtime + standardized MMA event schema. Useless without facility buy-in — politics, not code, is the wedge.

Six primitives every problem reaches for

1. Differentiable forward models
PSF, visibility, RT, lensing, waveforms, N-body. PyTorch/JAX prototypes exist; production codes are C/Fortran with no autograd.

2. Simulation-based inference (SBI)
Normalizing flows, neural posterior estimation. Each domain reinvents it independently.

3. GPU-native physics kernels
Gridding, FFT, tree-walks, opacity tables, ray-tracing, fluid solvers.

4. Streaming inference runtimes
Alert / visibility / event firehoses, deterministic latency, GPU residency.

5. Spatial-temporal indexing at scale
HEALPix-aware columnar formats, GPU spatial joins.

6. Foundation models for astronomy
AstroCLIP, AstroPT, light-curve transformers — trained over the same surveys, no shared substrate.

Six primitives. No shared C++ substrate. No LLVM-level IR.

Buildability filter — where can a small team ship?

Tier	#	Problem	Why this tier
1	7	Exoplanet retrieval	MB datasets, ExoJAX2 validates pattern, 1 GPU
1	9	Strong lens modeling	Euclid Q1 + HSC samples public, 1 GPU
1	3	Radio imaging	MeerKAT/VLA archives public, JAX prototypes exist
1	8	Catalog cross-match	All catalogs public, single-workstation prototype
1	1	LSST alert firehose	Stream live since Feb 2026, brokers in pain
2	4	N-body emulators	Cluster hours required for emulator training
2	6	GW parameter estimation	DINGO proves pattern; LSC adoption is political
3	2	SKA central signal proc.	EB/day-class hardware to validate
3	5	GRMHD surrogates	EHT collab guards the simulation library
3	10	Multi-messenger coord.	Useless without facility buy-in

Tier 1 = the buildable set. #7 has the smallest blast radius and the cleanest IR story.

Importance ranking — what's irrecoverably lost?

#1 LSST Alerts — most urgent
Time-perishable. A kilonova caught at +0.5 days is a different paper than one at +3 days. **The photons don't come back.** Everything else works on cached data.

#4 N-body — most fundamental
Euclid + Rubin + DESI + Roman (∼\$10B) are emulator-limited, not signal-limited. Determines whether dark energy gets pinned down or stays $\sigma \approx 0.02$.

#7 Exoplanet RT — most civilizational
Biosignature detection lives here. JWST data *being binned away today.* HWO (\$11B, 2040s) is bet on this being 100× better.

#8 Cross-match — highest leverage
Boring; quiet force multiplier. Every other problem eventually calls cross-match. Postgres of astronomy.

Most urgent: #1. **Most foundational:** #4. **Most civilizational:** #7. **Highest leverage:** #8.

The synthesis — what the substrate looks like

- **Front End**: Python bindings · DSL · natural-language goals
- **Astronomy IR**: spectra · visibilities · light curves · catalogs · events
- **Differentiable Physics**: RT · lensing · interferometry · waveforms · N-body (C++/CUDA)
- **Inference Engine**: amortized SBI · normalizing flows · autograd · MCMC fallback
- **Streaming + Catalog**: alert ingest · HEALPix indexing · GPU spatial join

Same shape as LLVM. Same shape as PyTorch. Missing in astronomy.

Why a C++ kernel with autograd is non-negotiable

1. **Throughput.** A single JWST retrieval needs 10^6–10^7 RT forward evaluations. Each one touches GB-scale opacity tables. Python wrappers cost 10–100× in the hot path.
2. **Determinism for reproducibility.** Bit-reproducible posteriors across hardware and years are required for biosignature claims.
3. **GPU residency.** Opacity tables, atmospheric layers, line lists — all want to stay on device. Python orchestration constantly bounces them through host memory.
4. **Composability.** The same RT primitive must be callable from MCMC, NF posterior, and gradient-based optimization without rewriting.
5. **Vendor longevity.** A telescope built in 2026 takes data until 2050+. The kernel that processes its photons must compile in 2050.

Python and Julia live *above* the kernel — for retrieval scripts, agent reasoning, UI. The hot path is C++.

Argus

A differentiable atmospheric retrieval kernel

“LLVM for the sky — starting with a single planet.”

One-line pitch. A C++20/CUDA core for differentiable line-by-line radiative transfer of exoplanet atmospheres, with an autograd-aware IR and a built-in simulation-based inference engine. Composable, GPU-resident, vendor-neutral.

What it replaces. The fragmentation of TauREx, POSEIDON, CHIMERA, petitRADTRANS, ExoJAX — five PhD codebases that don’t compose. Argus is the kernel; those become passes (or clients).

Why now. JWST is producing native-resolution spectra that are being binned away. ExoJAX2 (Apr 2025) proved differentiable RT works. Ariel launches 2029. The 24-month window to set the standard is open.

Argus — 18-month MVP roadmap

1. **M1–3 The IR.** Typed graphs of opacity sources, atmospheric layers, instrument PSFs, observational geometry. Serialization, content-addressed runs, provenance.
2. **M3–6 The kernel.** GPU-resident opacity tables (HITRAN/HITEMP/ExoMol). First differentiable transmission-spectrum solver in C++/CUDA. Validates against petitRADTRANS on the WASP-39b benchmark.
3. **M6–9 The inference engine.** Amortized SBI via normalizing flows. Compare against POSEIDON / CHIMERA on public JWST spectra: **target 10× speed, equal or better posteriors**, no binning required.
4. **M9–12 Lensing pass.** Add a strong-lensing forward model in the same IR. Proves the substrate generalizes beyond exoplanets.
5. **M12–18 Imaging pass.** Add a differentiable interferometric forward model. Three disjoint sub-fields running on one kernel = the substrate claim.

Zero physical hardware required. Public JWST + HSC + MeerKAT data; single-workstation MVP through M9.

Risks & mitigations

Upstream tool churn. Opacity databases (HITRAN, ExoMol), JWST pipeline, ExoJAX2 all evolve fast.
Mitigation: the IR stays stable; opacity sources are pluggable passes.

Adoption inertia. Retrieval community is small (∼50 active groups), each emotionally attached to a code.
Mitigation: ship a 10× speedup on a public benchmark first; let the speed sell itself. Open-source under Apache 2.0.

C++/CUDA hiring. Astronomy talent skews Python.
Mitigation: expose Python bindings from day one; recruit from HPC and games industry, not from astronomy.

IR over-design. Trying to be the LLVM of astronomy on day one is how you ship nothing.
Mitigation: ship the exoplanet wedge first, prove the IR *retroactively* by adding lensing and imaging passes in M9–18.

Sources

- Rubin Observatory — first alerts launch (Feb 2026), UW DiRAC announcement
- SKA — full-scale processing on Summit, exascale challenges (Phil. Trans. R. Soc.)
- N-body — generative-model challenge (CompAstroCosmo), large-scale dark matter sims review
- GRMHD — EHT 80M CPU-hr simulation library
- GW PE — AI for GW data analysis review, LIGO→LISA model-agnostic inference
- Exoplanet retrieval — VIRA (MNRAS 2024), ExoJAX2 (ApJ 2025)
- Cross-match — petabyte-era cross-match overview
- Lensing — LEMON (arXiv 2603, Mar 2026), first CNN lens analysis (Nature)
- Brokers — Fink LSST classifiers (A&A)
- Multi-messenger — Argonne RADAR
- Imaging — accelerated CLEAN (A&A 2026)
- SED emulators — Speculator (arXiv)

Argus

All-seeing.

The kernel that compiles photons into atmospheres.

Project repo: `~/Desktop/Future/Argus`

Brief: `Argus.tex` • Code: `include/argus` + `src/`